Exploring strategy learning in the "Tools Environment"

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Abstract

Humans are incredibly adept at learning rich contextual information about objects from only a few encounters. With respect to tool use, we are able to quickly master novel problem solving strategies, flexibly identifying how both familiar and unfamiliar objects can be employed to solve challenges confronting us. Here, we run a pilot set of experiments evaluating the "Tools Environment," a video game, as a candidate for computational study of the cognitive processes supporting strategy learning. Participants were asked to solve puzzles requiring the selection and placement of an object in an initially stable 2-D physical scene in order to accomplish goals such as getting an object past a set of obstacles or launching an object into a container. We find that, for three of the four tasks examined here, players' performance in the game improves in a structured manner, corresponding to measurable mastery of increasingly "successful" strategies as more trials are completed.

Keywords: tool use; object-based reasoning; intuitive physics

Introduction

Tool use is intimately related to higher level cognitive functions and psychological processes. Some developmental psychology studies have suggested that the first instances of tool use in human children may occur as early as before the end of the first year, establishing tools and the use of tools as a fundamental mode of representation learning and planning behavior (Lockman, 2000; McCarty, Clifton, & Collard, 2001; Keen, 2011). Though certain neurobiological processes supporting the acquisition of new representations via mastery of new tools have been investigated fairly extensively, the cognitive processes supporting the subtleties of tool use, and more broadly, human strategy learning and planning behavior, remain poorly characterized (Goldenberg & Spatt, 2009; Cardinali et al., 2009). What precisely are the cognitive mechanisms that endow us with our inexhaustible abilities to imagine novel ways to use both familiar and unfamiliar objects, develop general strategies for solving classes of problems, determine when and how strategies can be adapted to novel physical scenarios, and, generally, integrate representational reasoning with intuitive models of the physical world to iteratively compose objects under relational constraints? What is the richest way in which we can computationally describe and model these mechanisms?

In this paper, we attempt to take a first step towards answering these questions by evaluating the "Tools Environment," a set of physical action planning tasks, as a test-bed both for computational study of the aforementioned cognitive mechanisms supporting strategy learning and for the development of algorithms inspired by the most recent efforts in deep RL, with features adjusted to reflect experimentally uncovered insights. To that end, we ask subjects to solve a set of virtual physical tasks via the selection of an object from a set of options and its placement in a physical scene. We then use a finite mixture model to infer strategies employed by participants across experimental trials corresponding to each set of physical tasks. We find that though the results remain noisy, the inference model performs fairly well as a cursory analysis, showing positive strategy acquisition in participants over time across a subset of the trials studied here.

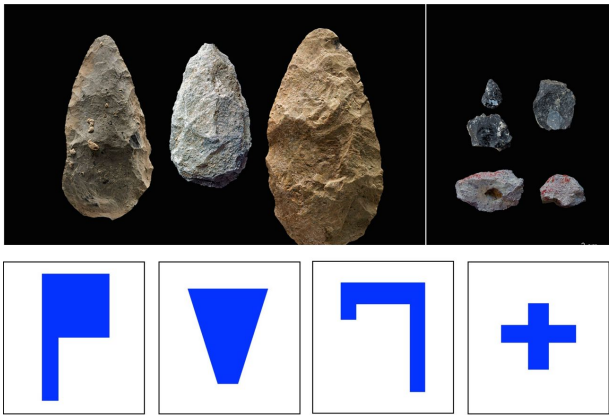

Figure 1: **Tools as insight to human behavioral and cognitive psychology.** Whether physical or virtual, we are remarkably skilled at recognizing new tools and new functionalities. In disciplines such as archaeology, developmental psychology, and anthropology, researchers have long used tools and tool use as markers for the development of key psychological processes in humans (Ambrose, 2001; Haidle, 2010; Lockman, 2000). *Top*: Neolithic tools from the Olorgesailie basin in Kenya dated to 320,000 y.a, currently recognized as the earliest evidence of human complex tool use and design (Potts et al., 2018). *Bottom*: 2-D objects from the "Tools Environment," described and discussed in proceeding sections of this paper, used by subjects to solve physical action planning tasks.

The Tools Environment

Participants

We recruited $N = 60$ participants on Amazon's Mechanical Turk using the psiTurk (Gureckis et al., 2016) experimental framework. Each participant was treated in accordance with the protocols of the MIT Committee on the Use of Humans as Experimental Subjects (COUHES).

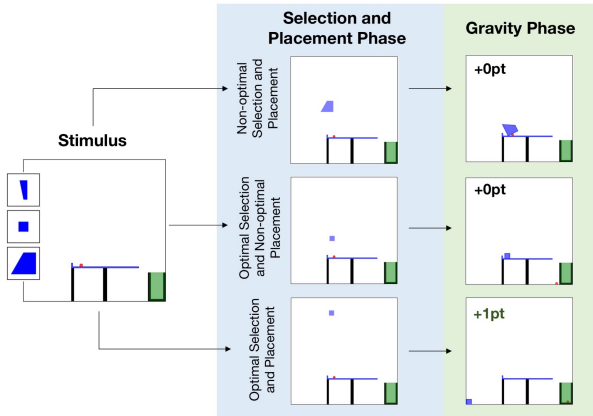

Figure 2: **The tools environment.** Given a stable scene, the task is to select a tool and place it in the scene in order accomplish a goal such as tipping an object over, or launching an object into a container (as in the *Catapult* task, shown above). Ultimately, the player must get the red object to end up in the green goal zone after stability is reached at the end of the "gravity phase". Three examples of performing the task in a single trial are presented here. *Top*: A non-optimal tool is selected and non-optimally placed (+0 points). *Middle row*: The optimal tool is selected, but is placed too low to gather enough momentum to successfully propel the ball into the goal location within the container (+0 points). *Bottom row*: The optimal tool is once again selected and is now placed optimally, successfully propelling the ball into the container (+1 point).

Stimuli and Design

The stimuli were composed of four 2-D, physical tasks (*Basic*, *Catapult*, *Table B*, and *Towers*) each requiring a different set of objectives, such as tipping an object over, knocking an object onto the floor, getting an object past a set of obstacles, or launching an object into a container, to be realized by the player or the agent. The principal objects of interest in each task were a red object (either a ball or block) as well as a green *goal zone*. Each task was also paired with a set of three external, convex objects, any single one of which could be selected by the player and used as a tool to ensure that at least one of the red objects ended up stable and within the green *goal zone* once the task was over, corresponding to a successful solution of the task.

By slightly perturbing parameters such as the shapes, sizes, heights, and widths of individual objects as well as mirror-flipping and rotating them, several hundred distinct trials were constructed for each task. The set of all trials was filtered to include only those in which a solution was possible and calibrated to ensure that random action would not reliably and constantly lead to a solution in any single trial over the full set of tasks. For the purposes of the human experiment, fifteen trials were randomly selected corresponding to each task.

During an initial *introductory* session, participants were first told to familiarize themselves with the physics engine and the general environment, with the requirement that they had to complete at minimum two actions over the course of at least 30 seconds. Participants were then given 900 seconds to complete two preliminary "practice trials," disjoint and dissimilar from any of the experimental trials. In the *experimental* session, the participants were first informed that although some trials might seem challenging, all were guaranteed to have a solution. Then, each participant was presented with one of four sets of fifteen trials, all trials within the set corresponding to a single, arbitrarily selected task.

The participants received points depending on whether they were able to solve each trial. One point was earned for every trial in which an optimal external object was successfully used by the participant as a tool to ensure that at least one red object ended up inside of the *goal zone*. If the participant failed to solve a trial after 120 seconds, unable to get a red object inside the *goal zone*, they were given the option to move on to the next trial, with no points either granted or deducted. The maximum possible scores for the *introductory* and *experimental* sessions were two and fifteen points, respectively.

Procedure

There were two phases in each trial: the *selection and placement phase* and the *gravity phase*. Each trial began in the *selection and placement* phase, during which a static scene and a set of three external objects were presented to the player. To use one of the external objects as a tool, the participant had to first select the object by clicking on it and then place it by clicking on a free region of the scene (i.e. a region unoccupied by any other objects). As soon as an object was successfully placed, the *gravity phase* was initiated and the physical interactions between all of the objects in the scene and the newly introduced tool were simulated with a Mujoco physics engine (Todorov et al., 2015). The *gravity phase* concluded once stability was once again reached by all of the objects.

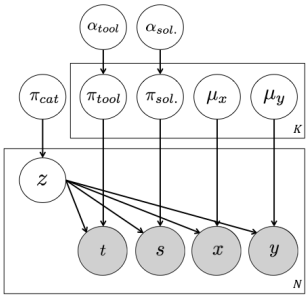

Figure 3: **Mixture model architecture.** In this paper, parameters were inferred from a symmetric Dirichlet distribution, so as not to give net bias towards any particular tool or solution state. Priors were set to be equal for the normalized placement coordinates, (x, y).

Strategy Inference Procedure

In an effort to logically decompose participants' actions across trials into distinct *strategies*, we used a hierarchical Bayesian finite mixture model such as in (Kemp, Perfors, & Tenenbaum, 2007) (see Fig. 3). For each trial, we considered the set of all observations to be composed of the tool chosen by the participant, the respective placement coordinates (x, y) of the tool, and whether this placement successfully solved the trial. Before inference, placement coordinates were normalized to $(\mu_x, \mu_y = 0,\ \sigma_x, \sigma_y = 1)$ and the number of global categories was set to $K = 4$, in an effort to prevent under-generalization. Parameters were then inferred according to the following specifications:

$$\pi_{cat} \sim \text{Dirichlet}(\alpha = [0.25, 0.25, 0.25, 0, 25])$$

$$\alpha_{tool}, \alpha_{shape} \sim \text{Exp}(a = 1)$$

$$\pi_{tool} \sim \text{Dirichlet}(\overline{\alpha} = [\alpha_{object_1}, \alpha_{object_2}, \alpha_{object_3}])$$

$$\pi_{sol.} \sim \text{Dirichlet}(\overline{\alpha} = [\alpha_{success}, \alpha_{failure}])$$

$$\mu_x, \mu_y \sim \text{Normal}(\mu = 0, \sigma = 2)$$

$$z_i \sim \text{Categorical}(\overline{p} = \overline{\pi}_{cat}, \overline{v} = [0, 1, 2, 3])$$

$$x \sim \text{Normal}(\mu = \mu_x, \sigma = 0.2)$$

$$y \sim \text{Normal}(\mu = \mu_y, \sigma = 0.2)$$

$$tool \sim \text{Categorical}(\overline{p} = \overline{\pi}_{tool}, \overline{v} = [\text{obj}_1, \text{obj}_2, \text{obj}_3])$$

$$solution \sim \text{Categorical}(\overline{p} = \overline{\pi}_{sol.}, \overline{v} = [\text{success}, \text{failure}])$$

Strategy inference over the data for each trial was conducted using Metropolis-Hastings Markov Chain Monte Carlo (MCMC) in the probabilistic programming language WebPPL (Goodman & Stuhlmüller, 2014) (see Fig. 4 for an example illustration).

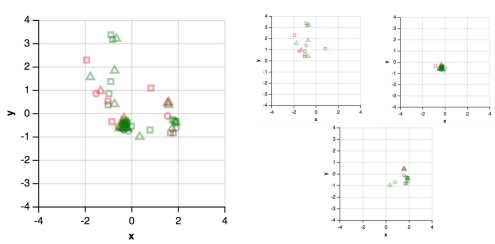

Figure 4: **Strategy inference.** Here we plot a decomposition of data corresponding to a single trial in the *Table B* task into three strategies. Each point above represents a placed object, its coordinates corresponding to the normalized placement coordinates (x, y) of the object, its shape representative of the object's enumeration in this particular trial, and its color symbolizing whether the placement was successful. (This figure was generated with WebPPL (Goodman & Stuhlmüller, 2014) visualization features.)

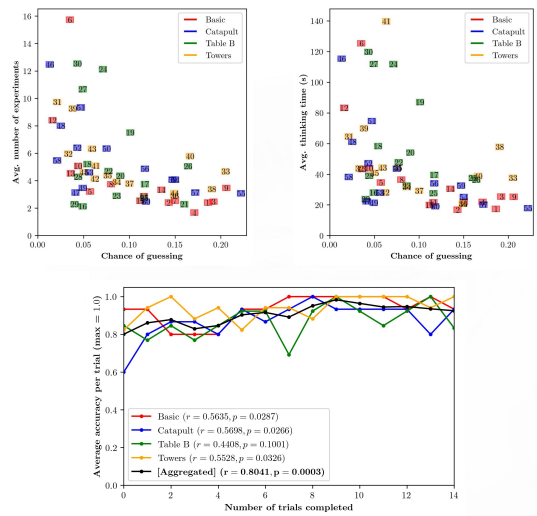

Figure 5: **Aggregated behavioral results.** We create plots as in (Dasgupta et al., 2018). *Top*: plot of trial difficulty (chance of random object selection and placement being successful) versus average number of practice placements made by participants (*left*) and average thinking time (*right*) for each trial. Points are labeled with trial indicators. *Bottom*: learning curves displaying average accuracy per trial versus number of trials completed.

Experimental Results

Behavioral Analysis

Overall, participants performed extremely well across all four tasks and consistently solved every trial more frequently than would be expected under chance performance (79.9% average accuracy versus 9.1% average accuracy by chance placement).

As in (Dasgupta et al., 2018), we find that across individuals, simulation and experimentation are tightly linked, with the average time participants spent on a trial correlated with the number of placements they made ($r(60) = 0.8543, p < 0.0001$). Participants were also sensitive to the difficulty of trials across tasks. As the probability of solving a trial by selecting and placing an object by chance decreased, participants spent more time thinking on average ($r(60) = -0.4750, p < 0.0001$ see Fig. 5, upper right) and made increased numbers of placements ($r(60) = -0.5262, p < 0.0001$ see Fig. 5, upper left). Notably, however, we find average solution rates to exhibit a low correlation with trial difficulty when assessed with this metric ($r(60) = 0.2571, p = 0.0473$). This result suggests that participants are sensitive to features of trial beyond the chance of random action's success.

We also observe a statistically significant correlation between average performance accuracy per trial and number of completed trials for all tasks, with the exception of *Table B* (see Fig. 5, bottom).

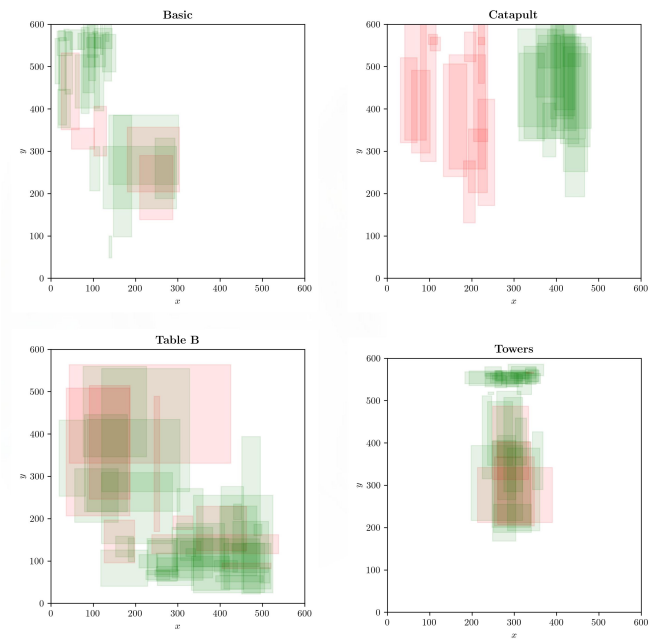

Figure 6: **Aggregated strategy results by task.** In each of the plots above, we show the strategies inferred from the human experimental data by the inference model across the fifteen trials corresponding to each task. Each plotted rectangle represents a single strategy, with center corresponding to average tool placement coordinates (μ_x, μ_y) calculated across data points attributed to the strategy, and width and height corresponding to the respective standard deviations of placement coordinates, (σ_x, σ_y). The color of each rectangle (red or green) is representative of whether attributed data points corresponded to a successful solution of the trial on average (green implying that $\geq 50\%$ of points attributed to the strategy were successes, red that $\geq 50\%$ were failures).

Strategy Analysis

Across tasks, we find that only on a trial-to-trial basis does the proposed inference model perform well with respect to producing descriptive strategy classifications. Comparing strategies inferred by the model across trials corresponding to a single task in order to deduce the features of each task shared by all trials, or to pinpoint gross, task-level, general strategies that a participant must learn in order to consistently achieve a high accuracy across trials, remains largely untenable. However, this model does serve fairly well for a cursory analysis of the tasks; as shown in Fig. 6, there does exist some measure of visibly identifiable global structure in the strategies deduced across trials for the *Catapult* and *Towers* tasks. With respect to the *Catapult* task, this structure takes the form of a horizontal division centered at $x = 300$ between the two largest clusters, one populated by successful strategies and the other by failing strategies (see Fig. 6, upper right). This division corresponds to the higher probability of solution given object placement in the right half of each *Catapult* trial, $x_{placement} \geq 300$. Respectively, in *Towers*, we see a cluster of relatively low σ_x and σ_y successful strategies centered at $\sim (300, 500)$ above a much larger, heterogeneous cluster of strategies centered around $(300, 300)$ (see Fig. 6, lower right). In this case, the smaller tighter cluster corresponds to the existence of a solution in the covered respective region across the majority of *Towers* trials.

As a final analysis of the strategy inference model, we superimpose the curves representing participants' strategy-calibrated accuracy per trial versus the number of trials completed onto the previously displayed accuracy per trial curves (see Fig. 6, bottom). The results are displayed in Fig. 7. Overall, we find that for all tasks with the exception of *Table B*, participants' average strategy-calibrated, accuracy per trial correlates slightly more strongly with the number of completed trials than face-value average accuracy ($r_{s,Basic}(15) = 0.61 > 0.56 = r_{Basic}(15), r_{s,Catapult}(15) = 0.64 > 0.57 = r_{Catapult}(15)$, and $r_{s,Towers}(15) = 0.70 > 0.55 = r_{Basic}(15)$). Note that we see higher increases in correlation in the *Catapult* and *Towers* tasks, where as discussed, our inference model yielded a visible degree of task-level strategy structure. In *Table B*, however, we see quite the opposite effect, with already low correlation dropping in strength substantially, once again reflecting the strategy heterogeneity across trials displayed in Fig. 6 (lower left) and the lack of significant positive correlation between accuracy and completed trials (as in Fig. 5, bottom).

In summary, the mixture model's performance across tasks indicates that some subset of the following is true: either 1.) the model captures a fair representation of strategies employed by players across trials corresponding to a given task, but priors on inference should be adjusted according to the

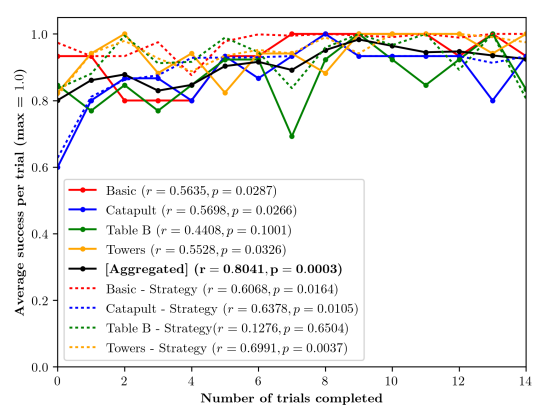

Figure 7: **Average strategy-calibrated accuracy and face-value accuracy versus completed trials.** Note that "strategy-calibrated accuracy" refers to the average accuracy of the strategies employed by participants throughout each trial, corresponding to a relative measure of how successful the strategy tends to be on average. Given a longer period of game-play, with participants having more time to experiment, we would expect that more successful strategies would be mastered.

specific features of a task to ensure that there is task-level structure across the strategies inferred from trial data, 2.) the model presents a fair representation of strategies and the presence of noise can be mostly attributed to this particular experiment's relatively small sample, or 3.) the model must be restructured.

Conclusions and Future Directions

In order to test which of the three aforementioned points are true and develop a fine-grained and accurate computational model of the cognitive mechanisms driving planning behavior and strategy learning in humans, it is fairly clear that more data, corresponding to a larger and more diverse set of tasks with greater numbers of participants as well as longer sequences of trials, must be gathered. In this experiment, with only fifteen trials and data from only fifteen participants studied for each task, sample size unquestionably played a role in the noise inherent in our analysis.

Overall, however, we see the positive strategy learning behavior that we observe here for the *Basic*, *Catapult*, and *Towers* tasks, despite the experiment's sample size, as fundamentally very promising - the "Tools Environment" is certainly a favorable candidate for further study and experimentation in the challenged domains.

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